

PA4 Weekly Report — Sprint 4 Task Tracking & Meeting Minutes

Freestyle Chess Mobile Web — Group 06

Course: CSC13112 - UI/UX Design **Lecturer:** Dr. Le Khanh Duy **Teaching Assistant / Instructor:** MSc. Pham Nguyen Son Tung **Class:** 23KTMP2 **Assignment:** Project Assignment 4 (PA4) — High-Fidelity Prototype & Summative User Study **Product Scope:** Freestyle Chess Mobile Website on Smartphone Browser **Sprint:** Sprint 4 (2-week sprint: August 24, 2026 – September 06, 2026) **Process Methodology:** RUP + Scrum **Repository:** <https://github.com/HB4305/TKGD-PA4>

1. Project & Team Information

1.1. Project Overview

Field	Details
Course	CSC13112 - UI/UX Design (FIT - HCMUS)
Assignment	Project Assignment 4 (PA4) — High-Fidelity Prototype & Summative User Study
Group / Class	Group 06 — Class 23KTMP2
Product Scope	Freestyle Chess Mobile Website on Smartphone Browser
Sprint	Sprint 4 (PA4)
Sprint Duration	2 weeks (August 24, 2026 – September 06, 2026)
Meeting Link	https://meet.google.com/yat-svzi-mkm
Live Interactive Demo	https://testpa4.vercel.app
Video Demo (YouTube)	https://www.youtube.com/watch?v=jkVPSIJTUtY

Sprint Goal Implement a fully interactive High-Fidelity Prototype using React + Vite + TypeScript based on the two winning solutions from PA3 (**Nav-1**: Fixed Bottom Nav & **Sch-2**: Date Strip + Dedicated 8×8 Match Detail). Conduct a Within-Subjects Counterbalanced Summative Usability Study with 6 participants (*P01* → *P06*) comparing the prototype against the legacy mobile web (freestylechess.com). Measure TCT, Success Rate, Errors (ERR-1/2/3), SUS benchmark, and Likert satisfaction. Build an interactive Swiss Modern presentation slide deck, record video proofs, and complete all required deliverables for PA4 submission.

1.2. Team Members & Scrum Roles

No.	Full Name	Student ID	Scrum Role & Primary Responsibilities	Contribution
1	Le Mai Hoai Bao	23127326	Product Owner / Frontend Developer - Lead React/Vite/TypeScript prototype development and deployment to Vercel.- Manage GitHub repository, coordinate 45° camera video proofs, and produce YouTube demo.	100%
2	Lam Huu Khanh	23127205	Scrum Master / UX Researcher - Design Summative Study framework, AB/BA order allocation (4/2 actual split).- Synthesize raw data logs, calculate mean TCT, SUS scores, Likert distributions, and statistical tests.	100%
3	Pham Chi Bao Ninh	23127446	UI Designer / Data Observer - Architect CSS Design Tokens (variables.css), ensure touch targets meet the project target of $\geq 48 \times 48$ px.- Serve as Observer & Note-taker, classify ERR-1/2/3 interaction errors during testing sessions.	100%
4	Truong Cong Thien Phú	23127455	Content Lead / Co-Facilitator - Populate realistic tournament/match datasets and design bilingual survey instruments.- Assist in conducting user testing sessions and lead post-study semi-structured interviews.	100%
5	Phung Ngoc Tuan	23127510	Lead Facilitator / Slide Lead - Guide participants through testing protocol, stopwatch timing, and participant logistics ($N = 6$).- Build interactive Swiss Modern HTML/CSS slide deck (PA4-assignment-briefing.html) and Speaker Script.	100%

2. Meeting Records Overview

In accordance with the RUP + Scrum course framework, Group 06 conducted 4 formal meetings during Sprint 4:

Meeting Record	Date & Time	Mode / Location	Purpose
Sprint Planning Meeting	August 25, 2026 · 21:00 – 22:30	Google Meet — link	Define Sprint 4 goals, translate PA3 winning solutions into technical WBS tasks, assign deadlines.
Weekly Scrum Meeting 1	August 30, 2026 · 21:00 – 22:30	Google Meet / Online — link	Review Week 1 progress: React prototype core screens, design system tokens, study protocol setup.
Weekly Scrum Meeting 2	September 03, 2026 · 21:00 – 22:30	Google Meet / Online — link	Review 6 user testing sessions ($P01 \rightarrow P06$), verify YouTube video proofs, review initial slide deck.

Meeting Record	Date & Time	Mode / Location	Purpose
Sprint Review & Retrospective	September 06, 2026 · 21:00 – 22:30	Google Meet — link	Evaluate deliverables vs PA4 criteria, analyze quantitative & qualitative results, finalize submission package.

3. Sprint Planning Meeting Minutes

- **Date & Time:** August 25, 2026 · 21:00 – 22:30
- **Location:** Online (Google Meet: <https://meet.google.com/yat-svzi-mkm>)
- **Members Present:** Le Mai Hoai Bao, Lam Huu Khanh, Pham Chi Bao Ninh, Truong Cong Thien Phu, Phung Ngoc Tuan (5/5)
- **Members Absent:** None

3.1. Sprint Goal Definition

Develop a production-grade, responsive High-Fidelity Prototype running smoothly on mobile browsers that embodies **Nav-1** and **Sch-2**. Execute a rigorous within-subjects summative usability study ($N = 6$) comparing baseline vs prototype across TCT, error rates, SUS, and Likert satisfaction. Package all findings into a high-impact interactive presentation deck and formal submission reports.

3.2. Work Breakdown Structure (WBS) & Task Allocation

Task ID	Task Description	Assignee	Est. Hours	Priority	Target Deliverable / Output
TSK-01	Setup React + TypeScript + Vite project repo, routing, and deploy to Vercel	Le Mai Hoai Bao	6h	High	Working Vercel URL, Repo skeleton
TSK-02	Design CSS design tokens, typography, dark/gold palette, and touch target rules	Pham Chi Bao Ninh	8h	High	<code>variables.css</code> , responsive foundation
TSK-03	Implement 5-tab fixed Bottom Navigation Bar (Nav-1) with active states	Le Mai Hoai Bao	8h	High	<code>BottomNav.tsx</code> component
TSK-04	Implement Date Strip carousel, Date Picker modal, and match filtering (Sch-2)	Bảo, Ninh	12h	High	<code>DateStrip.tsx</code> , <code>Schedule.tsx</code>
TSK-05	Implement Match Detail screen with interactive 8×8 SVG chessboard	Le Mai Hoai Bao	10h	High	<code>MatchDetail.tsx</code> , 8×8 board
TSK-06	Curate realistic tournament database (players, FIDE	Truong Cong	6h	Medium	<code>matches.ts</code> , player datasets

Task ID	Task Description	Assignee	Est. Hours	Priority	Target Deliverable / Output
	ratings, matches, boards)	Thien Phu			
TSK-07	Design Summative Usability Study protocol & AB/BA order allocation	Lam Huu Khanh	8h	High	06-PA4-SummativeUserStudy-Framework.md
TSK-08	Prepare bilingual questionnaires (Demographic, SUS 10-item, Likert 5-dimension)	Phu, Khanh	6h	High	DemographicForm.md, SUS.md, Likert.md
TSK-09	Recruit 6 participants (P01 → P06) with diverse backgrounds, schedule sessions	Phung Ngoc Tuan	6h	High	Participant roster & testing schedule
TSK-10	Setup Over-the-Shoulder camera recording rig and conduct dry-run	Bao, Tuan	4h	Medium	Standardized testing environment
TSK-11	Execute 6 Summative Usability Testing sessions (20 min/session)	Tuan, Phu, Ninh	12h	High	Raw stopwatch logs, observation notes
TSK-12	Synthesize quantitative metrics (TCT, Success Rate, ERR-1/2/3, SUS, Likert)	Lam Huu Khanh	10h	High	Statistical tables, verification logs
TSK-13	Process video recordings, upload 6 proof videos + 1 interactive demo to YouTube	Le Mai Hoai Bao	8h	High	6 YouTube proof links + Demo video
TSK-14	Build 17-slide interactive Swiss Modern presentation (PA4-assignment-briefing.html)	Phung Ngoc Tuan	16h	High	PA4-assignment-briefing.html
TSK-15	Author 18-minute Speaker Script for 4 presenters and Demo Operator	Tuấn, Khánh	8h	High	PA4-speaker-script-VI.md
TSK-16	Consolidate reports, verify checklist, export PDFs, package 06-PA4.zip	All Members	8h	High	06-PA4-HifiPrototype.md, 06-PA4-SummativeUserStudy.md, 06-PA4-WeeklyReport.md

4. Weekly Scrum Meeting Minutes

4.1. Weekly Scrum Meeting 1 (Week 1 Review)

- **Date & Time:** August 30, 2026 · 21:00 – 22:30
- **Location:** Online (Google Meet: <https://meet.google.com/yat-svzi-mkm>)
- **Members Present:** Le Mai Hoai Bao, Lam Huu Khanh, Pham Chi Bao Ninh, Truong Cong Thien Phu, Phung Ngoc Tuan (5/5)
- **Members Absent:** None

Status Reports by Member

1. Le Mai Hoai Bao (Product Owner / Developer):

- **Completed Tasks:** Built interactive prototype with Home, Schedule, and Match Detail screens; deployed live version to Vercel; integrated Bottom Navigation Bar (**Nav-1**) with haptic feedback transitions.
- **To-do Tasks:** Polish Date Strip animation; add 8×8 chessboard piece rendering for Carlsen vs Lê Quang Liêm; set up video recording apparatus.
- **Issues / Obstacles:** Ensuring the bottom navigation bar remains firmly pinned above mobile OS virtual home bars (iOS Home Indicator & Android gesture pill). Resolved via `env(safe-area-inset-bottom)`.

2. Lam Huu Khanh (Scrum Master / UX Researcher):

- **Completed Tasks:** Established experimental framework: Within-subjects AB/BA design with an actual final allocation of $4 \times AB + 2 \times BA = 6$ users; defined 4 testable hypotheses (H_1 to H_4); created raw logging template.
- **To-do Tasks:** Monitor testing data collection; calculate Brooke's 1996 SUS formula and paired delta comparisons once logs are submitted.
- **Issues / Obstacles:** Ensuring wash-out interval (90 s) between System 1 and System 2 is strictly adhered to prevent carryover fatigue.

3. Pham Chi Bao Ninh (UI Designer / Observer):

- **Completed Tasks:** Finalized CSS design tokens in `variables.css`; verified that all touch targets satisfy $\geq 48 \times 48$ px with ≥ 8 px spacing; developed error categorization taxonomy (ERR-1 Slip, ERR-2 Mistake, ERR-3 Hesitation).
- **To-do Tasks:** Act as Observer during live testing; log timestamps and error codes in real-time.
- **Issues / Obstacles:** Distinguishing between ERR-1 (touch slip) and ERR-3 (hesitation) during rapid user finger movements. Standardized definition: hesitation is > 3 s pause without touching.

4. Truong Cong Thien Phu (Content Lead / Co-Facilitator):

- **Completed Tasks:** Curated realistic match data from FIDE Freestyle Chess 2026; authored bilingual survey sheets (Demographic, SUS 10-item, 5-dimension Likert).
- **To-do Tasks:** Assist Lead Facilitator during test sessions; conduct post-test qualitative interview.
- **Issues / Obstacles:** Making sure survey questions in Vietnamese accurately convey the nuances of standard English SUS phrasing.

5. Phung Ngoc Tuan (Lead Facilitator / Slide Lead):

- **Completed Tasks:** Recruited 6 participants ($P01 \rightarrow P06$) with balanced profiles (handedness, chess experience, mobile habits); completed test rehearsal dry-run.

- **To-do Tasks:** Facilitate 6 user testing sessions on August 31 – September 02; begin constructing presentation slide structure.
- **Issues / Obstacles:** Coordinating participant time slots across evening hours to ensure zero external noise interruptions during video recording.

Actions & Summary

- Prototype deployed successfully on Vercel (testpa4.vercel.app).
- Full testing setup approved; schedule locked for 6 sessions across 2 days.

4.2. Weekly Scrum Meeting 2 (Week 2 Review & Study Synthesis)

- **Date & Time:** September 03, 2026 · 21:00 – 22:30
- **Location:** Online (Google Meet: <https://meet.google.com/yat-svzi-mkm>)
- **Members Present:** Le Mai Hoai Bao, Lam Huu Khanh, Pham Chi Bao Ninh, Truong Cong Thien Phu, Phung Ngoc Tuan (5/5)
- **Members Absent:** None

Status Reports by Member

1. Le Mai Hoai Bao (Product Owner / Developer):

- **Completed Tasks:** Recorded over-the-shoulder video proofs for 6 sessions; uploaded videos to YouTube; recorded official 4-minute interactive Hi-fi Demo video ([jkVPS1JTUyY](https://www.youtube.com/watch?v=jkVPS1JTUyY)).
- **To-do Tasks:** Compile deliverable document [06-PA4-HifiPrototype.md](#); package final ZIP.
- **Issues / Obstacles:** Hand occlusion on camera for left-handed angle; adjusted camera position to 45° right elevation.

2. Lam Huu Khanh (Scrum Master / UX Researcher):

- **Completed Tasks:** Completed full quantitative data synthesis ($N = 6$): Total TCT reduced by 77.7% (118.6 s $\rightarrow$ 26.5 s), SUS increased from 33.3 (**Grade F**) to 92.5 (**Grade A+**), Slip errors eliminated (3.5 $\rightarrow$ 0.0); direct-success and error rates are reported descriptively because $N = 6$.
- **To-do Tasks:** Complete [06-PA4-SummativeUserStudy.md](#); verify raw logs against video timestamps.
- **Issues / Obstacles:** None; continuous measures were synthesized quantitatively, while categorical success/error results are reported descriptively because $N = 6$.

3. Pham Chi Bao Ninh (UI Designer / Observer):

- **Completed Tasks:** Consolidated error logs: 0 ERR-1 slips observed on Prototype vs 21 total slips on Baseline; captured in-situ photos from testing sessions for presentation evidence.
- **To-do Tasks:** Format visual cards and photo galleries for slide deck.
- **Issues / Obstacles:** None.

4. Truong Cong Thien Phu (Content Lead / Co-Facilitator):

- **Completed Tasks:** Transcribed and coded qualitative interviews into 3 core thematic clusters: (1) One-handed reachability, (2) Day-by-day schedule browsing, (3) Match card information density.
- **To-do Tasks:** Draft future iteration design implications (auto-month threshold, direct date input).

- **Issues / Obstacles:** None.

5. Phung Ngoc Tuan (Lead Facilitator / Slide Lead):

- **Completed Tasks:** Completed all 6 test sessions (*P01* → *P06*); developed 17-slide interactive HTML deck ([PA4-assignment-briefing.html](#)) with embedded raw log modal, comparative bar charts, and 6 participant photo cards; authored 18-minute 4-speaker script.
- **To-do Tasks:** Conduct full dry-run rehearsal with all 4 speakers and demo operator.
- **Issues / Obstacles:** Slide 12 card grid layout required fine-tuning to fit 6 participant photos side-by-side gracefully at 1920×1080 resolution. Resolved via 6-column CSS grid.

Actions & Summary

- All 6 user testing sessions successfully completed and verified.
- Rehearsal scheduled for September 05; submission deadline set for September 06.

5. Weekly Task Tracking Table

Week	Member	Assigned Task IDs & Details	Output / Evidence Artifact	Status	Notes & Role
W1	Le Mai Hoai Bao	TSK-01, TSK-03, TSK-05: Setup React/Vite repo, build Bottom Nav & Match Detail 8×8 board.	Codebase, Vercel deployment	Done	Frontend & PO
W1	Lam Huu Khanh	TSK-07, TSK-08: Design Summative study protocol, AB/BA order matrix, survey questionnaires.	SummativeUserStudy-Framework.md	Done	UX Research Lead
W1	Pham Chi Bao Ninh	TSK-02: Design CSS tokens (variables.css), verify touch target boundaries ≥ 48 px.	variables.css , Error taxonomy	Done	UI & Observation Lead
W1	Truong Cong Thien Phu	TSK-06, TSK-08: Curate match/tournament dataset, prepare bilingual survey forms.	Dataset files, Survey forms	Done	Content & Survey Lead
W1	Phung Ngoc Tuan	TSK-09, TSK-10: Recruit 6 participants, setup 45° camera rig, conduct testing dry-run.	Testing schedule & lab setup	Done	Facilitation Lead
W2	Le Mai Hoai Bao	TSK-10, TSK-13, TSK-16: Record 6 video proofs, record 4-min interactive demo, package ZIP.	6 YouTube proofs + Demo video	Done	Video Lead & Submitter

Week	Member	Assigned Task IDs & Details	Output / Evidence Artifact	Status	Notes & Role
W2	Lam Huu Khanh	TSK-12, TSK-15, TSK-16: Analyze quantitative metrics (TCT/SUS/Likert), co-author speaker script.	Quantitative log tables, Script	Done	Data Analyst & Script
W2	Pham Chi Bao Ninh	TSK-11, TSK-12: Observe 6 test sessions, log ERR-1/2/3 codes, prepare photo evidence.	Error log sheet, Photo assets	Done	Observer & Evidence
W2	Truong Cong Thien Phu	TSK-11, TSK-12: Co-facilitate testing, transcribe post-test interviews, qualitative thematic coding.	06-PA4-ParticipantsReplies.md	Done	Interviewer & Qualitative
W2	Phung Ngoc Tuan	TSK-11, TSK-14, TSK-15: Facilitate 6 sessions, build 17-slide HTML deck, lead speaker script.	PA4-assignment-briefing.html	Done	Slide Lead & Facilitator

6. Git Commit Tracking & Traceability Table

The table below maps representative repository commits to their corresponding tasks, dates, and authors:

Commit Hash	Date & Time (ISO 8601)	Author	Task ID	Commit Message / Description	Linked Artifacts
27e2bfa	2026-08-27 14:15:20 +0700	Le Mai Hoai Bao	TSK-04	feat: enhance DateStrip component with new date picker and smooth scrolling animations	DateStrip.tsx
3fb6c5c	2026-08-28 10:22:14 +0700	Le Mai Hoai Bao	TSK-04	feat: Add place holder for matches with no game	Schedule.tsx
92a7b97	2026-08-28 16:40:05 +0700	Truong Cong Thien Phu	TSK-06	feat: add matches from FIDE Freestyle Chess World Championship 2026 to dataset	matches.ts
e1a0c53	2026-08-29 11:18:42 +0700	Le Mai Hoai Bao	TSK-01	feat: update more data and change some UI	Prototype UI assets

Commit Hash	Date & Time (ISO 8601)	Author	Task ID	Commit Message / Description	Linked Artifacts
ca395b2	2026-08-29 18:30:11 +0700	Le Mai Hoai Bao	TSK-03	feat: add Rating page	Rating.tsx
c857636	2026-08-30 09:45:18 +0700	Phung Ngoc Tuan	TSK-15	Add speaker script for PA4 presentation on Freestyle Chess Mobile Web	PA4-speaker-script-VI.md
dd33430	2026-08-30 15:20:00 +0700	Pham Chi Bao Ninh	TSK-02	add paper prototype images	Reference assets
5d01c66	2026-08-30 20:12:35 +0700	Truong Cong Thien Phu	TSK-08	feat: add demographic form, Likert satisfaction scale, and SUS questionnaire for user testing	Survey documents
3845a2a	2026-08-30 21:05:40 +0700	Truong Cong Thien Phu	TSK-08	docs: add interview guide for user testing feedback	InterviewGuide.md
6a5d058	2026-08-31 08:30:12 +0700	Lam Huu Khanh	TSK-07	feat: add user study framework documentation	SummativeUserStudy-Framework.md
097d389	2026-08-31 10:15:00 +0700	Truong Cong Thien Phu	TSK-08	docs: rename demographic form, Likert satisfaction scale, and SUS questionnaire	Survey naming standardization
89a846f	2026-09-01 14:22:18 +0700	Le Mai Hoai Bao	TSK-01	Refactor code structure for improved readability and maintainability	Codebase refactor
4591c15	2026-09-02 11:05:33 +0700	Phung Ngoc Tuan	TSK-14	fix: adjust iframe image styling for better display	Slide UI components
28ec814	2026-09-02	Phung Ngoc	TSK-14	docs: add agenda in slide	Presentation deck structure

Commit Hash	Date & Time (ISO 8601)	Author	Task ID	Commit Message / Description	Linked Artifacts
	16:40:12 +0700	Tuan			
11cbe45	2026-09-03 09:30:25 +0700	Lam Huu Khanh	TSK-12	docs: add detailed verification testing documentation	06-PA4-VerifyTesting.md
9fce968	2026-09-03 14:10:00 +0700	Truong Cong Thien Phu	TSK-11	Add file to manage test participants results	Logging framework
d3c5376	2026-09-03 18:25:40 +0700	Truong Cong Thien Phu	TSK-11	Add data analysis for P01, P02, P03, P05, P06	Participant reply logs
92ef7e5	2026-09-04 10:12:00 +0700	Truong Cong Thien Phu	TSK-11	docs: add participant 06 reply and analysis	06-PA4-ParticipantsReplies.md
fec9374	2026-09-04 16:50:30 +0700	Phung Ngoc Tuan	TSK-14	feat: add presentation slides, project documentation, and user study assets	Slide deck & photo galleries
c9f8bcb	2026-09-05 11:20:15 +0700	Le Mai Hoai Bao	TSK-16	feat: update slide and rename file for submission	Submission structure
557ca00	2026-09-06 14:18:22 +0700	Phung Ngoc Tuan	TSK-13	fix: update video link for Participant P06 in replies document	06-PA4-ParticipantsReplies.md
3ee4c16	2026-09-06 14:20:00 +0700	Le Mai Hoai Bao	TSK-16	Merge branch 'main' of https://github.com/HB4305/TKGD-PA4	Branch synchronization

7. Sprint Review & Deliverables Evaluation

7.1. Deliverables Checklist vs. PA4 Assignment Requirements

Assignment Requirement (PA4 Spec)	Group 06 Deliverable & Evidence	Status
Requirement 1: High-Fidelity Prototype (55%) — Mobile web prototype implemented with high design fidelity; resolves PA2/PA3 user problems (Nav-1 & Sch-2); fully interactive on mobile browsers; demo video provided.	Responsive React + Vite web application deployed on Vercel: https://testpa4.vercel.app . Three complete screens, YouTube demo, and architecture documentation delivered in 06-PA4-HifiPrototype.pdf .	100% Completed
Requirement 2: Summative Usability Study (40%) — Within-subjects AB/BA study with $N \geq 5$; standardized protocol; quantitative metrics; qualitative feedback; video proof.	Complete report in 06-PA4-SummativeUserStudy.pdf , 6 sessions, actual order allocation 4 AB / 2 BA, and 6 YouTube proof links. Direct-success and error results are reported descriptively for the small sample.	100% Completed
Requirement 3: Live Demo & Presentation (<i>supporting deliverable; not separately weighted in Section C</i>) — 15–20 minute presentation and live prototype run.	17-slide presentation deck and 18-minute speaker script.	100% Completed
Requirement 4: Weekly Report (5%) — Complete task tracking, sprint meetings, retrospective, commit traceability and contribution breakdown.	Full sprint documentation, meeting minutes, and commit mapping in 06-PA4-WeeklyReport.md / 06-PA4-WeeklyReport.pdf .	100% Completed

7.2. Summative Usability Study Key Outcomes

- **Hypothesis H_1 (Efficiency / TCT Reduction):**
 - **Task 1 (One-Handed Navigation):** TCT decreased from 36.3 s to 11.8 s (**giảm 67.5%**, $p < 0.001$).
 - **Task 2 (Schedule & Match Detail):** TCT decreased from 82.3 s to 14.7 s (**giảm 82.1%**, $p < 0.001$).
 - **Total Time-on-Task:** Decreased by 77.7% overall (118.6 s $\rightarrow$ 26.5 s).
- **Hypothesis H_2 (SUS Benchmark):**
 - SUS Score increased by +59.2 points, from 33.3/100 (**Grade F**) on Baseline to 92.5/100 (**Grade A+**) on Prototype, far exceeding the industry average benchmark of 68.0.
- **Hypothesis H_3 (Effectiveness & Error Reduction):**
 - For **Task 2**, Direct Task Success reached 100% on Prototype versus 50% on Baseline; Task 1 baseline direct success was 67%.
 - Slip Errors (ERR-1) were completely eliminated (0.0 errors vs 3.5 errors/user on Baseline) due to large touch targets ($\geq 48 \times 48$ px) and thumb-zone optimization.
- **Hypothesis H_4 (Subjective Satisfaction):**
 - 5-dimension Likert score reached 4.7/5.0 (vs 2.9/5.0 on Baseline), confirming overwhelming user preference for the redesigned interface.

7.3. Member Contribution Summary

Team Member	Student ID	Primary Roles & Key Contributions	Completion Quality	Contribution Score
Le Mai Hoai Bao	23127326	Product Owner, React/Vite development, Vercel deployment, YouTube video proofs & demo production, repository governance.	Excellent	100%
Lam Huu Khanh	23127205	Scrum Master, Summative study experimental design, AB/BA order allocation, statistical synthesis (TCT/SUS/Likert), report writing.	Excellent	100%
Pham Chi Bao Ninh	23127446	UI Designer, CSS design tokens, project touch-target target, live test observer & error classification.	Excellent	100%
Truong Cong Thien Phu	23127455	Content Lead, match/tournament dataset curation, survey questionnaires, co-facilitation, qualitative thematic synthesis.	Excellent	100%
Phung Ngoc Tuan	23127510	Testing Lead, participant logistics ($N = 6$), lead test facilitator, Swiss Modern HTML presentation deck, speaker script.	Excellent	100%

8. Sprint Retrospective Meeting Minutes

- **Date & Time:** September 06, 2026 · 21:00 – 22:30
- **Location:** Online (Google Meet: <https://meet.google.com/yat-svzi-mkm>)
- **Members Present:** All 5 members present (5/5)

8.1. What Went Well

- **Production-Grade Hi-fi Prototype:** Building the prototype in React + TypeScript with realistic animations and haptic-like visual states provided participants with a truly authentic app experience that felt native.
- **Counterbalancing:** The AB/BA allocation reduced carryover effects, but the final distribution was 4/2 rather than perfectly balanced; this is documented as a study limitation.
- **Interactive Presentation Engineering:** The self-contained Swiss Modern presentation ([PA4-assignment-briefing.html](#)) with embedded raw log modal, comparative SVG charts, and in-situ photo evidence created an exceptionally compelling visual narrative for the briefing.

8.2. What Went Wrong & Problems Encountered

- **Mobile Screen Glare under Outdoor Lighting:** Session P02 (outdoor cafe test) experienced sunlight reflection on the camera lens.
 - *Root Cause:* Direct overhead midday sunlight at the cafe outdoor terrace.
 - *Correction:* Adjusted camera exposure and rotated seating angle by 45° to eliminate reflection.
- **Video Timestamp Cross-Referencing Overhead:** Manually verifying think-aloud quotes and micro-hesitations against 6 multi-gigabyte video files required substantial time.

- *Root Cause:* Note-taker initial timestamps were recorded via wristwatch rather than synced stopwatch.
- *Correction:* Re-synced all video audio tracks to central stopwatch timestamps during final data consolidation.

8.3. What Can Be Done Differently in Future Projects

- **Automated Telemetry Logging:** Embed lightweight client-side event logging (e.g., tap coordinates, time-per-screen) directly into the prototype code to automatically capture micro-timings alongside manual stopwatch logs.
- **Automated Monthly Boundary Scrolling:** As suggested by P01 and P03, implement auto-month header snapping when scrolling across date boundaries in future production releases.

8.4. Key Lessons Learned

- Summative usability testing with empirical rigor (counterbalancing, within-subjects, formal hypothesis testing) provides definitive, undeniable proof of design improvements that subjective opinions cannot refute.
- Investing effort into clean CSS tokens and component-driven architecture during early sprints pays massive dividends when adapting and polishing the final product.

9. Issues, Adjustments & Final Status

Issue Identified	Impact on Project	Corrective Adjustment Made
Mobile browser navigation bars occluded bottom tabs on certain Android viewports.	Risk of touch target clipping for Bottom Nav.	Added <code>padding-bottom: max(16px, env(safe-area-inset-bottom))</code> to keep tabs in prime Thumb Zone.
Baseline site lacked search and date filtering, causing participant P06 to exceed 120s timeout.	Skewed baseline time-on-task variance.	Recorded baseline completion as 120.0 s+ (Timeout/Assist) according to standardized assistance protocol.
Slide 12 gallery required displaying 6 participant photos side-by-side without vertical clipping.	Risk of overcrowded visual cards on 16:9 canvas.	Implemented responsive 6-column CSS grid with card hover micro-animations and compact typography.

Final Status

All Sprint 4 tasks for Project Assignment 4 (PA4) were completed 100% on schedule. All 5 team members maintained perfect attendance across all 4 sprint meetings and contributed equally to all deliverables. All 3 required submission files ([06-PA4-HifiPrototype.md](#), [06-PA4-SummativeUserStudy.md](#), [06-PA4-WeeklyReport.md](#)) have been verified, cross-checked against rubric criteria, and prepared for final packaging into [06-PA4.zip](#).